Initial base costs				
Incorporate SAW requirements into CISP			89,000	27,000
Develop detailed design specifications for SAWs			56,200	47,000
Provide Participants with detailed design specifications			3,000	
Develop automated monitoring systems			52,400	
Total base initial costs			**200,600**	**74,000**
Annual Base Costs				
Maintain and monitor CISP SAW requirements			56,600	
Maintain detailed design specifications			48,300	
Additional costs for third party annual audit			150,000	
Maintain automated monitoring systems and monitor			605,300	
Total base annual costs			**860,200**	
Additional Costs for All Participants in SAWs				
Initial				
Technical development costs	39,500,000			
Evaluate nine SAWs for compliance			167,000	
SAW operations implementation costs	21,700,000			
Total Additional Initial Costs	**61,200,000**		**167,000**	
Annual				
SAW usage costs		12,900,000		
Technical maintenance costs	19,000,000			
Total Annual Additional Costs	**19,000,000**	**12,900,000**		

Additional Costs for All Participants Excepted			
Additional Initial Costs	**1,289,600**	**2,250,000**	**1,048,800**
Additional Ongoing Costs	**417,400**	**2,250,000**	**1,160,100**

a. SAW versus Exception Decisions

Under the proposed amendments, each Participant will be required to limit some of its use of CAT Data to SAWs provided by the Plan Processor unless it obtains an exception to certain SAW usage requirements.[632] Consequently, each Participant will likely meet its regulatory obligations using one or more of three approaches. First, the Participant may decide to use the Plan Processor provided SAWs that would be established under the proposed amendments. Second, the Participant may decide to apply for an exception to allow it to use a different analytic environment to access and analyze CAT Data. Third, the Participant may decide to employ a 17d-2 or RSA to discharge its regulatory responsibilities. Each of these potential approaches has direct and indirect costs to the Participant that are discussed below.

In the first approach, a Participant may elect to use a SAW provided by the Plan Processor. The costs of operating and maintaining this SAW would be paid by the Participant, and the magnitude of these costs would be dependent on the resources used by the Participant within the SAW.[633] If a Participant adopts this approach, it may have lower expenses associated with maintaining its private analytic environment. However, to the degree that the Participant currently uses IT resources that it also uses for operational activities to perform its regulatory

632 Participants will be able to use the online direct query tool from their own analytic environments under certain restrictions, but the number of records of CAT Data they extract, and their access to Customer Information, would be limited for this manner of access. See supra Part II.C and Part II.D.

633 The estimated costs of SAWs are discussed further below.

activities, this may create inefficiencies because those resources may be less utilized during hours when operational demands are lower, such as when exchanges are not operating, if it performs regulatory activities in the SAW. Under this approach, to the degree that the lack of excess operational resources limit the Participant's ability to perform its regulatory activities in-house, the Participant may be able to insource more of its regulatory activities when working in the SAW, reducing its dependence on and costs associated with 17d-2s and RSAs.[634] Utilizing a SAW may also open competitive opportunities to the Participant to perform regulatory services for other Participants within its SAW.[635] Moving regulatory activities to the SAW is likely to entail significant implementation costs: the Participant would need to develop or license analytic tools for that environment or adapt its current analytical tools to that environment, and train its regulatory staff in using the SAW environment. The Commission preliminarily believes this approach is more likely to be adopted by Participants in Participant Groups that operate multiple exchanges because these costs might be spread over more exchanges,[636] and by Participants that already have a significant cloud presence because their implementation costs would likely be lower than those for a Participant that did not have a cloud presence.

In the second approach, a Participant may apply to use a private analytical environment through the exception procedure. In this approach, the Participant would incur costs to document that its private analytic environment meets the security requirements of the proposed amendments, and to adapt its analytic tools to those requirements. Further, the Participant would

---

634 RSAs are discussed further below.

635 See infra Part IV.B.1.

636 Participants that operate multiple exchanges often have commonalities in data structures and rules across their exchanges that allow economies of scale in performing regulatory activities.

incur costs associated with applying for and obtaining the exception, and complying with annual renewal requirements. The Participant may also encounter certain inefficiencies in accessing CAT Data to the extent that download speeds between the Central Repository and the private analytic environment are inferior to those within the SAW.[637] A Participant that adopts this approach may also choose to change the scope of its use of 17d-2s and RSAs as a provider or user of regulatory services through such agreements. For example, a Participant may choose to pursue an exception to the SAW use requirement and add additional 17d-2 and RSA coverage for functions that are more difficult to perform within its private analytic environment. Alternatively, there may be analytic tools that are more efficient to use outside of SAWs, allowing a Participant to provide regulatory services to other Participants that would be less efficient to provide in the SAWs. The Commission preliminarily believes this approach is more likely to be adopted by Participants that have a significant investment in private analytic workspaces, and proprietary tools for regulatory activities that are optimized for those workspaces.

In the third approach, a Participant would change its use of RSAs and 17d-2 agreements to avoid using a SAW or obtaining an exception to the SAW use requirement. This approach is likely to increase a Participant's expenses associated with RSAs and 17d-2 agreements, but may allow a Participant to avoid SAW expenses entirely. It is possible that even with maximal use of RSAs and 17d-2 agreements, a Participant may want to perform some regulatory functions that would not be possible with only use of the online targeted query tool. In this case, a minimal SAW would also have to be supported if the Participant did not wish to seek an exception to the SAW use requirement. The Commission preliminarily believes that this approach is most likely

---

637 See infra Part IV.B.2.

to be adopted by Participants that operate a single venue, and Participants that currently outsource much of their regulatory activities to other Participants. The Commission recognizes it is possible that many Participants will take this approach considering that many Participants make broad use of RSAs and 17d-2 agreements to discharge their regulatory responsibilities.

Finally, the Commission recognizes that a Participant may take a mixed approach to this decision. A Participant may elect to use the SAW for some regulatory activities, and outsource other activities that would significantly increase its use of resources in the SAW, and thus its costs of using the SAW. It is also possible that a Participant may choose to invest heavily in the SAW to compete in the market for regulatory services as an RSA provider, while also obtaining an exception to the SAW use requirement to allow it to capitalize on its current infrastructure.

b. Amendments for SAWs

The Commission is proposing amendments to the CAT NMS Plan that will require (1) the provision of SAW accounts; (2) data access and extraction policies and procedures, including SAW usage requirements; (3) security controls, policies, and procedures for SAWs; (4) implementation and operational requirements for SAWs.[638] The Commission preliminarily believes that the proposed amendments may improve the security of CAT Data in two ways.

First, to the extent that CISP security controls are implemented more uniformly than they would be under the CAT NMS Plan, security may improve by reducing variability in security control implementation.[639] Currently, each Participant would be responsible for implementing security controls in their analytic environments and their approaches are likely to vary if each

---

638 See supra Part II.C.

639 See supra Part II.C.3.

Participant designs those implementations to accommodate their current operations and analytic environments. This variability might result in some environments being more secure than others.[640] To the extent that having the Plan Processor provide SAWs that implement common security controls reduces this variability,[641] these provisions may increase CAT Data security by preventing relatively weaker implementations. The Commission recognizes it is also possible that the Plan Processor's implementation might be relatively less secure than an implementation designed by an individual Participant under the current CAT NMS Plan. The Commission preliminarily believes these provisions should improve security by reducing the variability of implementations as long as the Plan Processor's implementation of common security controls is relatively secure compared to other possible approaches. Further, the Commission preliminarily believes that the requirement that the Plan Processor must evaluate and notify the Operating Committee that each Participant's SAW has achieved compliance with the detailed design specifications before that SAW may connect to the Central Repository will further increase uniformity of security control implementations.[642]

Second, the proposed amendments may increase the uniformity of security monitoring across all environments from which CAT Data is accessed and analyzed.[643] By assigning this

---

640 The Commission preliminarily believes that different environments that satisfy the CISP might vary in their overall level of security due to differences in implementation, third-party software and policies and procedures for monitoring the security of the environments. To the extent that a bad actor would focus an incursion attempt upon the least secure environment, reducing variability between environments may improve CAT Data security by reducing vulnerabilities within environments from where CAT Data is accessed and analyzed.

641 See supra Part II.C.1.

642 See supra Part II.C.4

643 See supra Part II.C.4.

duty to a single entity, the Plan Processor, and making provisions for the uniformity of this monitoring through detailed design specifications, the proposed amendments may enhance the security of CAT Data by ensuring that security monitoring is uniform. Currently under the CAT NMS Plan, most security monitoring of environments other than the Central Repository would fall to the Participants that controlled those environments.[644] To the extent that the rigor of this monitoring and the manner in which requirements were implemented varied across Participants and the Plan Processor, some environments might be more robustly monitored than others, potentially delaying the identification of security issues within less robustly monitored environments. In addition, having a single entity perform this security monitoring may improve its quality by facilitating development of expertise of the single entity performing the monitoring. To the extent that the Security Working Group participates in the development of this monitoring, expertise from the wider group of Participants might also improve the quality of monitoring. Further, the Commission preliminarily believes that standardizing implementation of security protocols through the common detailed design specifications may be more efficient than having each Participant that implements a SAW or private environment for CAT Data do so independently because it avoids duplication of effort. This may also improve efficiency by reducing the complexity of security monitoring of environments from which CAT Data is accessed and analyzed because the detailed design specifications will include provisions that facilitate this central monitoring.

Finally, the Commission preliminarily believes that provisions of the proposed amendments that establish security controls, policies, and procedures for SAWs may improve CAT Data security. Currently, under the CAT NMS Plan, Participants must establish security

---

644 See supra Part IV.A.3.a.

protocols comparable to those required for the central repository for all environments from which Participants access CAT Data.[645] The proposed amendments require that SAWs comply with the same security standards as the Central Repository, including compliance with and common implementation of certain NIST SP 800-53 security controls, policies, and procedures. To the extent that the security controls, policies and procedures required for SAWs in the proposed amendments are more rigorous than what the Participants would implement under the current CAT NMS Plan, the security of CAT Data may be improved.

Table 3 summarizes the Commission's preliminarily cost estimates if all Participants were to work within SAWs. The Commission estimates that Participants would collectively incur $61.2MM in initial costs and $31.9MM[646] in ongoing annual costs, while the Plan Processor would incur $441,600[647] in initial costs and $860,200 in ongoing annual costs. These costs are discussed further in the analysis that follows.

Table 3:

Costs for all Participants to use SAWs ($)

	Participants		Plan Processor	
Activity	**Labor**	**External**	**Labor**	**External**
Initial				
Incorporate SAW requirements into CISP			89,000	27,000
Develop detailed design specifications for SAWs			56,200	47,000
Provide Participants with detailed design specifications			3,000	
Evaluate nine SAWs for compliance			167,000	
Technical development costs	39,500,000			
Develop automated monitoring system			52,400	
SAW operations implementation costs	21,700,000			

645 See supra text accompanying note 623.

646 ($19,000,000 + $12,900,000) = $31,900,000.

647 ($367,600 + $74,000) = $441,600.

Total initial costs	**61,200,000**		**367,600**	**74,000**
Annual				
Maintain and monitor CISP SAW requirements			56,600	
Maintain detailed design specifications			48,300	
Maintain automated monitoring system and monitor			605,300	
Additional costs for third party annual audit			150,000	
Technical maintenance of SAWs	19,000,000			
SAW usage costs		12,900,000		
Total ongoing costs	**19,000,000**	**12,900,000**	**860,200**	

Under the proposed amendments, the Plan Processor would be required to incorporate SAW-specific additions into the CISP.[648] The Commission preliminarily estimates the Plan Processor will incur approximately $89,000[649] in initial labor and $27,000[650] in external consulting costs to fulfill this requirement. The Commission preliminarily estimates the Plan Processor will also incur $56,600[651] in recurring annual costs to meet those provisions.

The Commission preliminarily estimates that the Plan Processor will incur initial, one-time costs of approximately $56,200[652] in labor costs and $47,000[653] in external legal and consulting costs to develop detailed design specifications for the technical implementation of the access, monitoring and other controls required for SAWs.[654] The Commission preliminarily

---

648 See supra Part III.D.3.a.

649 See supra note 498.

650 See supra Part III.D.3.a.

651 See supra note 501.

652 See supra note 503.

653 See supra Part III.D.3.a.

654 Id.

believes the Plan Processor will incur $3,000[655] in labor costs to make the required detailed design specifications available to the Participants, and will incur an additional $48,300[656] per year to maintain those detailed design specifications.

For the Plan Processor to evaluate each Participant Group's[657] SAW to confirm that the SAW has achieved compliance with the detailed design specifications and to notify the Operating Committee, the Commission preliminarily estimates that the Plan Processor would incur an initial, one-time expense of approximately $167,000.[658]

For the Plan Processor to build automated systems that will enable monitoring of the SAWs and Excepted Environments, the Commission preliminarily estimates that the Plan Processor would incur an initial, one-time expense of $52,400.[659] For the Plan Processor to maintain such systems and to monitor each Participant's SAW in accordance with the detailed design specifications, the Commission preliminarily estimates the Plan Processor would incur annual recurring costs of $605,300.[660] For each instance of non-compliance with the CISP or detailed design specifications, the Plan Processor would incur costs of $500 to notify the non-compliant Participant.[661]

---

655 See supra note 506.

656 See supra note 508.

657 The Commission preliminarily believes that each Participant Group will contract for a single SAW because it preliminarily believes that each Participant Group largely centralizes its regulatory functions that would require CAT Data.

658 See supra note 509. $18,550 per group x 9 groups = $166,950.

659 See supra note 510.

660 See supra note 511.

661 See supra note 512.